Kubernetes Observability Lab

Monitoring, alerting and hands-on labs for DSYS601 students

Kabir Thukral | Bachelor of Information & Communication Technologies

Ara Institute of Canterbury | Semester 2, 2026 | kabirthukral2@gmail.com

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Ara rau, taumata rau

Project partner: Ara DSYS601 teaching team

Scan to open the lab on GitHub

1 Introduction

DSYS601 students learn to run apps on Kubernetes. Seeing what is happening inside a cluster, and being told when something breaks, is a core DevOps skill, but there was no ready-made lab to teach it.

2 Aim

Build a reproducible Kubernetes lab with monitoring, dashboards and alerts that tutors can rebuild each cohort and students can learn from.

3 Method

I worked through 15 tasks from the industry brief, one at a time. Each had an evidence report, reviewed and reworked on feedback.

Build 3-node cluster with Kind

Deploy 3-tier demo app

Monitor Prometheus + Grafana (Helm)

Alert 4 custom alert rules

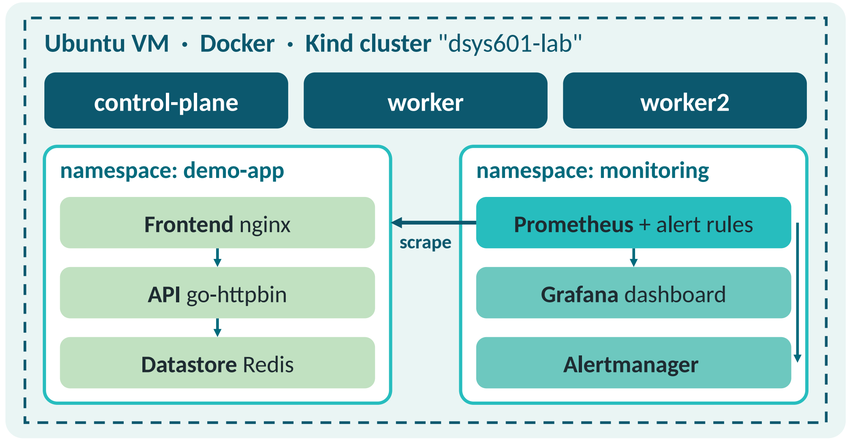
Break failure tests fire alerts

Hand over pinned versions + student labs

4 Outcomes

A complete lab rebuilt from one GitHub repository, plus six guided student labs (Lab 0 to Lab 5).

Figure 1. *Lab architecture*



3 cluster nodes

4 custom alerts

6 student labs

15 brief tasks

Figure 2. *Breaking things on purpose: each failure test fired its alert*

Failure test	DSYS601 alert	Result
Container keeps crashing	PodCrashLooping	Fired ✓
Memory held near its limit	ContainerMemoryNearLimit	Fired ✓
Broken readiness probe	PodNotReady	Fired ✓
Node under memory pressure	NodeUnderPressure	Fired ✓

Tools used

Ubuntu · VMware · Docker · Kubernetes · Helm · Prometheus · Grafana · Git · GitHub

5 Learnings

- Kubernetes in depth: pods, probes, resource limits and node pressure
- PromQL and alert design: tuning wait times to stop false alarms
- Reproducibility: pinned versions, image digests and rebuild tests
- Writing for others and acting on supervisor feedback

6 Conclusion

Tutors can rebuild a working, monitored cluster from one repo each cohort, and students learn by breaking things safely and watching the alerts fire.

References

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Acknowledgements

Course Convenor: Dr David Weir
david.weir@ara.ac.nz
Academic Supervisor: Dr Bernard Otinpong
bernard.otinpong@ara.ac.nz
Industry Supervisor: Abdul Rehman
abdul.rehman@ara.ac.nz